Brewster Angle Reflectometry (BAR) for Pyrosil® coating characterization on glass

INNOVENT



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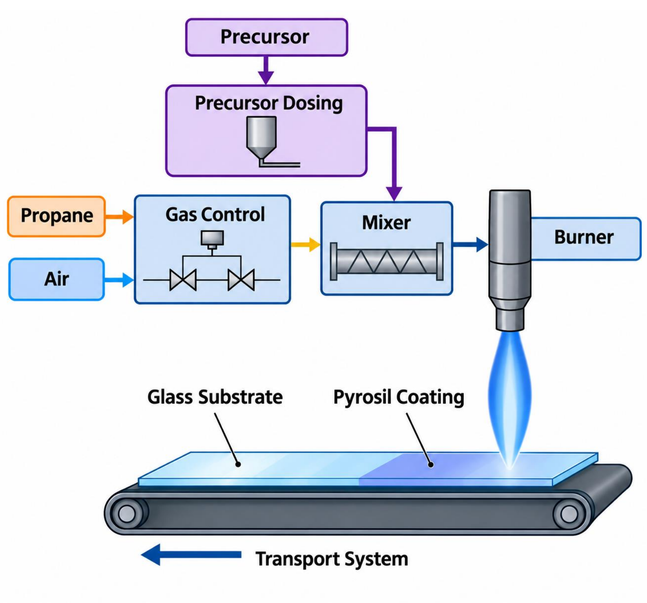
Motivation

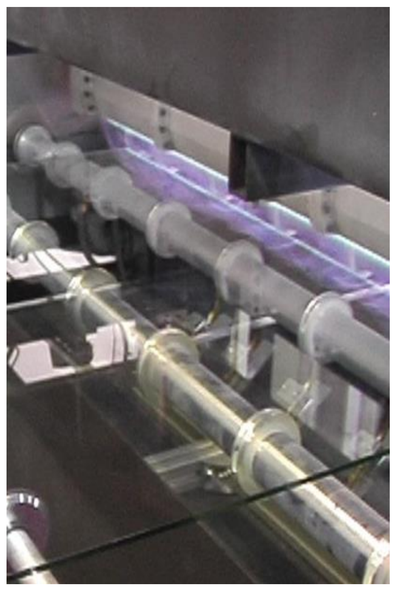
- coating and temper processes visualize previously invisible surface modifications on glass (corrosion, clouds, mechanical inprints, contaminations, cleaning marks)
- develop a coating to avoid or reduce aging of glass surfaces

Aim

- measure and control the Pyrosil® coating process of large area glass panes by Brewster Angle Reflectometry
- achieve a cost effective production of Pyrosil® coating on glass with defined optical coating parameters

Pyrosil® coating technology





Technical Data of Pyrosil® coating

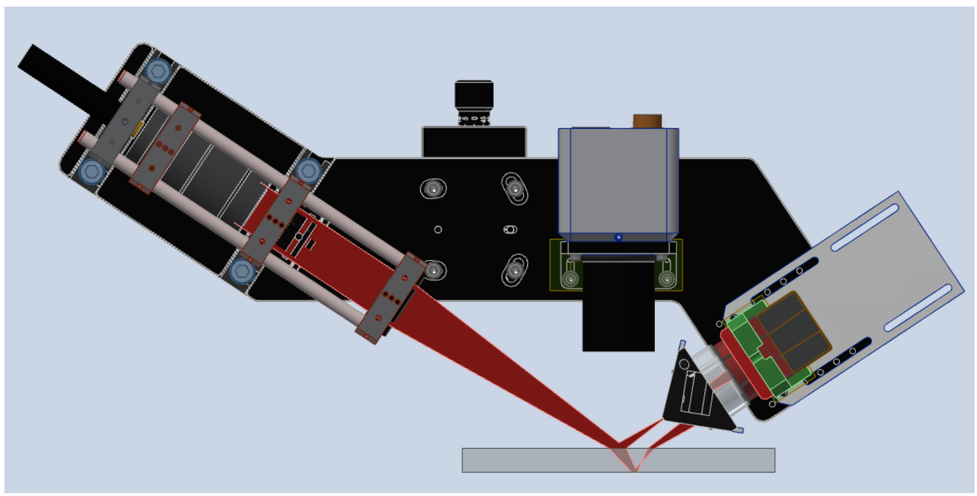
- Atmospheric pressure process
- Coating thickness: 5-100 nm with a roughness between 1-10 nm
- Available at 1.20 m scale

Key benefits

- Enhanced Durability: Protects glass from scratches, abrasion and corrosion
- Optical Clarity: Maintains high transparency and minimizes light reflection
- Easy-to-clean: Reduces surface contamination by facilitating the easy removal of dirt and water

Brewster Angle Reflectometry

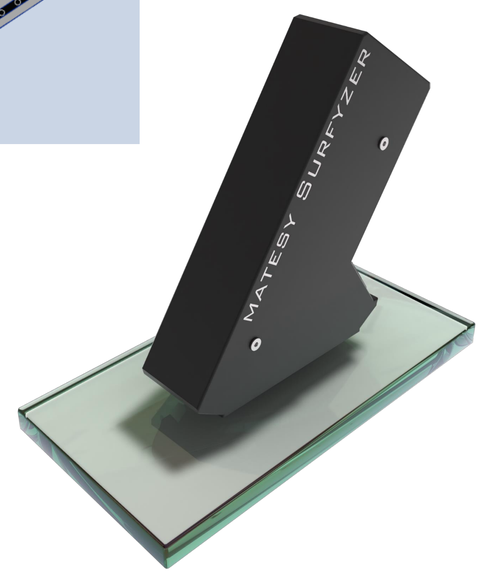
Camera measurement of reflectance curve with laser line illumination



Technical Data & Specifications

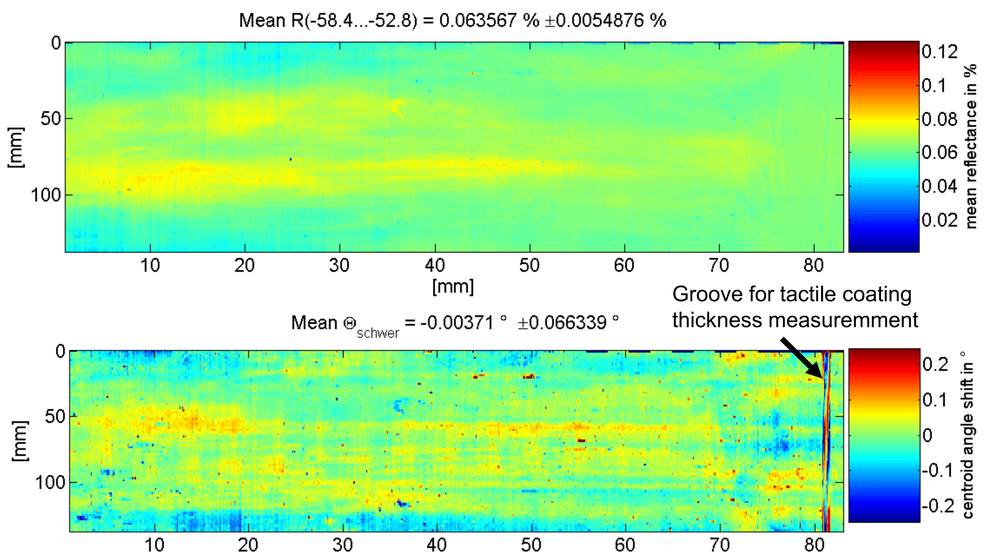
• Angle of incidence	56.5° ± 4°
• Scanning laser line	0.5 mm x 4 mm
• Scanning speed	≤ 250 mm/s
• Lateral resolution	≥ 0.25 mm x 0.125 mm
• Film thickness*	10 nm – 100 nm ±1 nm
• Film refractive index*	1.3 – 1.7 ± 0.01
• Laser illumination class	3R

* Within the validity range of Effective Medium Theory



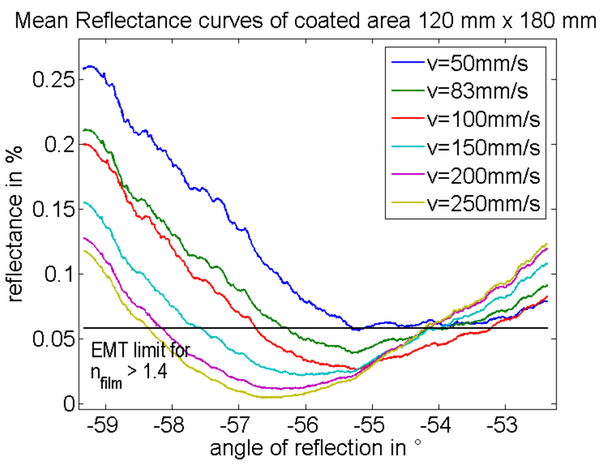
Measurement results

Optical parameters for coating inhomogenity visualization



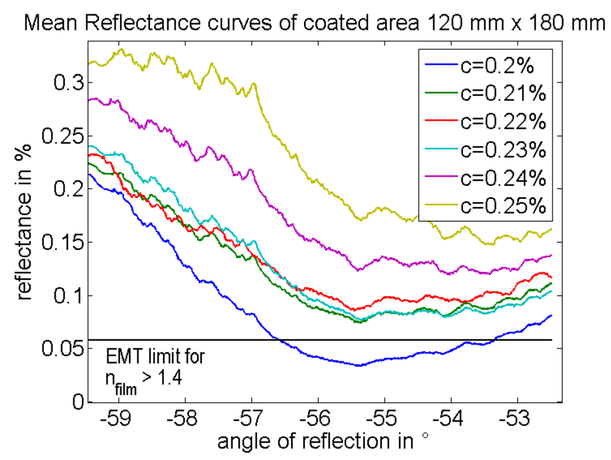
Change in reflectance curves as a function of coating speed

(4 runs, precursor concentration c = 0.2%)



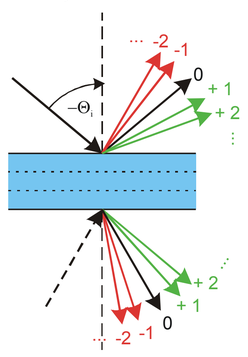
Change in reflectance curves as a function of precursor concentration

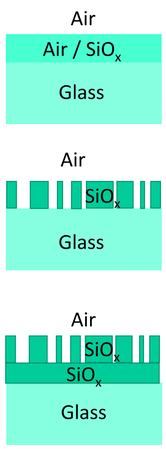
(4 runs, coating speed v = 83 mm/s)



Multi-layer stack calculation

Rigorous **C**oupled **W**ave **A**pproach
(calculated by UNIGIT https://unigit.net)



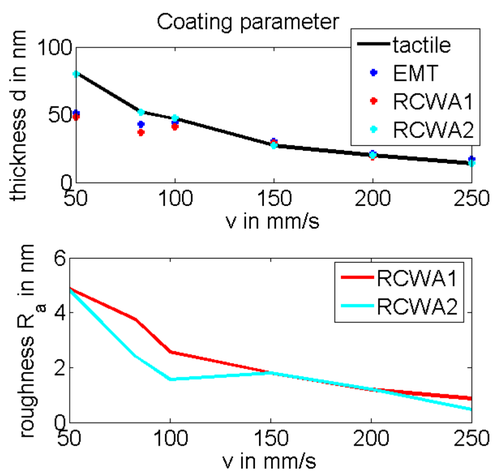


Effective Medium Theory
Homogenious layer by averaging n composite materials (no roughness)

RCWA one slice hard transition
Random distribution of air and SiO_x in one slice (roughness modulated)

RCWA two slice hard transition
Random distribution of air and SiO_x in two slices (roughness modulated)

Coating thickness and roughness as a function of coating speed



Conclusion

BAR
→ fast and sensitive coating analysis

Pyrosil®
→ compact base layer with rough upper layer

increased coating speed
→ decreasing film thickness
→ decreasing roughness

increased precursor concentration
→ increasing roughness

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